

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: ATOMIC & LASER PHYSICS

TRINITY TERM 2024

Friday, 14 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. Describe the physical origin of the spin-orbit interaction and derive an expression for it in an atomic system with a single electron (Thomas precession can be taken into account by the replacement $g_s \rightarrow g_s - 1$ for the g -factor of the electron, and need not be discussed). Show that configurations are split into two levels separated by

$$\Delta E = \frac{Z^4 \alpha^2 \hbar c R_\infty}{n^3 l(l+1)} ,$$

for $l \neq 0$, where l and n are the orbital angular momentum and principal quantum numbers respectively. [8]

Show that the following transitions have similar wavelengths:

- i) the $n = 2$ to $n' = 3$ transition in atomic hydrogen (H), and
- ii) the $n = 4$ to $n' = 6$ transition in the helium ion, He^+ .

For both, give the values of the quantum numbers n, l for which the largest fine-structure splitting occurs. Evaluate these splittings, expressing your answer as a fraction of the transition frequency in each case. State what other effect can cause a wavelength difference between these two atomic species and estimate this shift as a fraction of the transition frequency. [4]

Draw an energy level diagram (not to scale) for the $n = 2$ to $n' = 3$ transition in atomic hydrogen, labelled with appropriate quantum numbers; indicate any fine-structure splitting and all the allowed electric dipole transitions between these two shells ($|\Delta n| = 1$). List the relevant selection rules and give a physical justification for one of them. [5]

For both transitions, estimate the Doppler broadening in a gas at room temperature, expressing your answers as a fraction of the transition frequency. Suggest another line broadening mechanism that may occur when these lines are observed in the emission from an electrical discharge lamp that excites the atoms, or ions. [4]

For what value of atomic number Z does a hydrogen-like ion have a fine-structure splitting that is comparable with the energy of the $n = 2$ to $n' = 3$ transition in atomic hydrogen? State how the fine-structure splitting increases with atomic number Z for (neutral) alkali metal atoms, and justify this trend. [4]

$$\left[\begin{array}{l} \text{The Bohr magneton is } \mu_B = \frac{e\hbar}{2m_e} = \frac{1}{2} \alpha c e a_0 . \text{ For hydrogenic systems} \\ \left\langle \frac{1}{r^3} \right\rangle = \frac{2}{l(l+1)(2l+1)} \left(\frac{Z}{na_0} \right)^3 \text{ where } Z \text{ is the atomic number.} \end{array} \right]$$

2. For an atom in an external magnetic flux density $\mathbf{B}$, the interactions involving the spin angular momentum of the electron and nucleus have the form

$$\mathcal{H} = A\mathbf{I} \cdot \mathbf{J} + g_J\mu_B\mathbf{J} \cdot \mathbf{B} - g_I\mu_N\mathbf{I} \cdot \mathbf{B},$$

What is the physical origin of each of these three interactions? Explain the terms ‘strong’ and ‘weak’ field in the context of hyperfine structure, with reference to the relevant terms in the Hamiltonian. [5]

The hydrogen atom has a hyperfine splitting of 1420 MHz in the ground electronic configuration, and nuclear spin $I = 1/2$. Estimate the value of the magnetic flux density at which the crossover between the weak- and strong-field regimes occur. What quantum numbers are appropriate to label the eigenstates in these regimes? [4]

Find an expression for the eigenenergies in the weak-field regime, in terms of A, g_J, g_I and other relevant quantities. [5]

Find expressions for the four eigenenergies in the strong field regime. [5]

A collimated atomic beam of atoms, within a vacuum chamber, passes through a region with $B_z = 1 \text{ T}$ and a gradient of

$$\frac{\partial B_z}{\partial z} = 20 \text{ T m}^{-1}.$$

Show that atoms in the two upper high-field eigenstates experience similar acceleration and estimate its approximate value. Find the difference in the force on atoms in these two eigenstates, expressing your answer as a fraction of the mean value. Atoms in these two selected high-field eigenstates then travel into a low-field region where they are collected in a container that has walls coated with ‘non-stick’ material such that the hyperfine level does not change when they bounce off the walls. Sketch all four eigenenergies as a function of magnetic flux density, $|\mathbf{B}|$, showing how high- and low-field eigenstates connect. Hence, or otherwise, suggest how this system can be used for microwave amplification by stimulated emission of radiation (a hydrogen MASER). [6]

3. Explain why there is a large difference in the threshold pump power required to achieve laser oscillation for three-level and four-level lasers. [4]

In a laser medium the population densities of the two levels are N_2 and N_1 , their statistical weights are g_2 and g_1 , and ω_0 is the angular frequency of the transition. Starting from the Einstein treatment of radiation, derive an expression for the gain coefficient stating clearly any assumptions used. Show that the gain is proportional to the population inversion density $N^* = N_2 - g_2 N_1 / g_1$. [9]

Ions (of ytterbium) in a glass optical fibre have levels at energies E :

Level	E/hc [m^{-1}]
3	1 100 000
2	1 026 000
1B	106 000
1A	60 000
0	0

You may assume that the degeneracies of all the levels are unity. There is fast relaxation between levels 3 and 2 (upper band), and also in the lower band: from 1A to 0, and 1B to 0. A ytterbium-doped optical fibre has a radius of $2\text{ }\mu\text{m}$. Pump radiation at a wavelength of 910 nm propagates along the fibre and 86% of it is absorbed over the length $L = 0.5\text{ m}$ of the fibre; the absorption cross section is $\sigma_{\text{abs}} = 6 \times 10^{-25}\text{ m}^2$. This laser medium can provide gain at two wavelengths corresponding to transitions from upper level 2 to the lower levels 1A and 1B; both laser transitions have emission cross-section, $\sigma_{\text{em}} = 2.5 \times 10^{-25}\text{ m}^2$, at their respective wavelengths.

Calculate the number density of the ions. Find the fraction of ions in each of the levels 1A and 1B, assuming thermal equilibrium at temperature $T = 300\text{ K}$. Calculate the threshold pump energies, $E_{\text{A,th}}$ and $E_{\text{B,th}}$, that must be absorbed by the active ions for pulsed operation on the two laser transitions. For a pumping pulse of energy $E = 5E_{\text{B,th}}$ calculate the (unsaturated) round trip gain for the possible laser transition(s). [8]

The pump radiation is provided by a semiconductor diode laser which has an electrical-to-optical conversion efficiency of 60% . What is the maximum theoretical efficiency of the fibre laser? Comment on loss processes that reduce efficiency in practice. [4]

[The coefficients in the Einstein treatment of radiation are related by:

$$A_{21} = \frac{\hbar\omega_0^3}{\pi^2 c^3} B_{21} \text{ and } g_1 B_{12} = g_2 B_{21}.]$$

4. The radiative transfer equation of the form

$$\frac{dI_\nu}{dz} = -\alpha_\nu I_\nu + j_\nu ,$$

describes the change in specific intensity I_ν of radiation at frequency ν along a ray from absorptive and emissive processes, where α_ν is the specific absorption coefficient corrected for stimulated emission and j_ν is the specific emission coefficient. Show that by a suitable change of variables this can be written as

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu .$$

For a source function S_ν that is constant throughout the medium, show that

$$I_\nu = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu}) .$$

[6]

An apparatus is set up in which black-body radiation can pass through a flame of burning gas (considered to be a cloud of hot carbon particles) onto a detector whose output signal is proportional to the specific intensity I_ν . The positions of the black-body emitter, flame and detector remain fixed while measurements are recorded for the following three cases:

I_A for radiation from the black body transmitted through the flame,

$I_B = I_\nu(0)$ for radiation from the black body with the flame turned off,

I_C for radiation emitted from the flame alone, with the black body radiation blocked (or switched off).

These measurements are in the ratio $I_A : I_B : I_C = 0.85 : 1 : 0.15$. Find the value of τ_ν for the flame. Find an expression for the ratio $S_\nu/I_\nu(0)$ in terms of the ratios $X = I_A/I_B$ and $Y = I_C/I_B$, and evaluate it for this case.

[8]

For a homogeneously broadened transition of angular frequency ω_0 between a lower and upper level of degeneracies g_1 and g_2 respectively, the optical absorption cross section may be written as

$$\sigma_{\text{abs}} = \frac{g_2}{g_1} \cdot \frac{\pi^2 c^2 A_{21}}{\omega_0^2} \cdot g(\omega - \omega_0).$$

The upper level decays only by spontaneous emission to the lower level at a rate A_{21} , and $g(\omega - \omega_0)$ is a normalised line-shape function. Show that the resonant cross section is independent of the strength of transition. Evaluate the peak cross section for the 2s to 2p transition of lithium at a wavelength of 671 nm. Estimate the number density for which a cloud of lithium atoms with a diameter of 1 cm is optically thick. Comment on the circumstances for which purely lifetime broadened transitions may be observed.

[6]

Radiation from a source passes through a lithium vapour into a spectrograph that records the specific intensity over a range of wavelengths. What properties would make the source suitable to record the series of absorption lines, 2s–np, in lithium?

[5]

[The ionisation energy of lithium is 5.39 eV.]